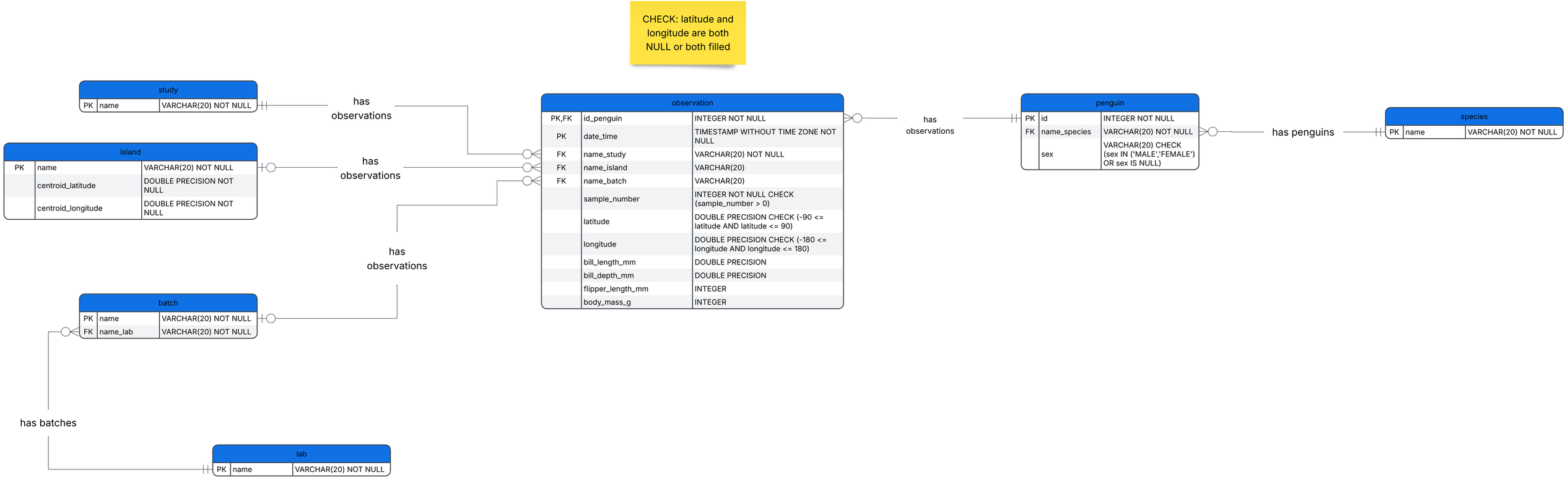
CHECK: latitude and longitude are both NULL or both filled
study
PK name VARCHAR(20) NOT NULL
has observations
island
PK name VARCHAR(20) NOT NULL
centroid_latitude DOUBLE PRECISION NOT NULL
centroid_longitude DOUBLE PRECISION NOT NULL
has observations
has observations
observation
PK,FK id_penguin INTEGER NOT NULL
PK date_time TIMESTAMP WITHOUT TIME ZONE NOT NULL
FK name_study VARCHAR(20) NOT NULL
FK name_island VARCHAR(20)
FK name_batch VARCHAR(20)
sample_number INTEGER NOT NULL CHECK (sample_number > 0)
latitude DOUBLE PRECISION CHECK (-90 <= latitude AND latitude <= 90)
longitude DOUBLE PRECISION CHECK (-180 <= longitude AND longitude <= 180)
bill_length_mm DOUBLE PRECISION
bill_depth_mm DOUBLE PRECISION
flipper_length_mm INTEGER
body_mass_g INTEGER
has observations
penguin
PK id INTEGER NOT NULL
FK name_species VARCHAR(20) NOT NULL
sex VARCHAR(20) CHECK (sex IN ('MALE','FEMALE') OR sex IS NULL)
has penguins
species
PK name VARCHAR(20) NOT NULL
batch
PK name VARCHAR(20) NOT NULL
FK name_lab VARCHAR(20) NOT NULL
has batches
lab
PK name VARCHAR(20) NOT NULL